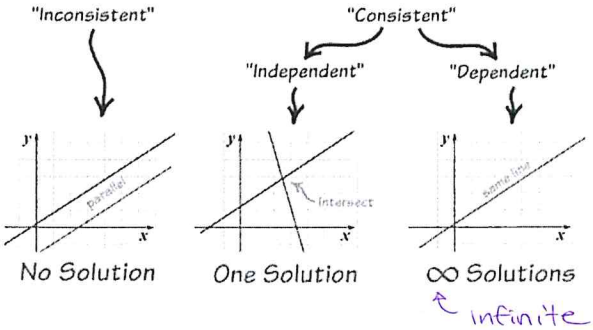
Intersection of Two Lines

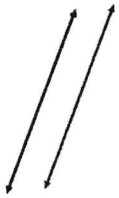
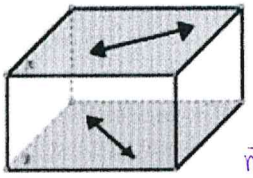
Recall: In R^2, solution(s) to a linear system, graphically, are point(s) of intersection.



A linear system is said to be consistent if it has at least one solution.

Otherwise, the system is said to be inconsistent.

In R^3, two lines can intersect in the following ways:

One Solution	Infinite Solutions	No Solution	No Solution
			 $\vec{m_1} \not\parallel \vec{m_2}$
Intersecting Lines $\vec{m_1} \not\parallel \vec{m_2}$	Coincident Lines $\vec{m_1} \parallel \vec{m_2}$	Parallel and Distinct Lines $\vec{m_1} \parallel \vec{m_2}$	Skew Lines *(Planes containing the lines are parallel, but the lines themselves are not)*

To solve a system algebraically, we can use the methods of substitution/elimination.

In R^2, given Cartesian equations ($Ax + By + C = 0$), we can simply use substitution or elimination.

If we were given vector equations…

Example 1: Determine the point of intersection between the two lines:

$$\vec{r}_1 = (-1,1,0) + t(3,4,-2), t \in \Re$$
$$\vec{r}_2 = (-1,0,-7) + s(2,3,1), s \in \Re$$

$\vec{m_1} = (3,4,-2)$, $\vec{m_2} = (2,3,1)$ — not collinear ∴ either intersecting or skew lines

Line #1: $x = -1 + 3t$, $y = 1 + 4t$, $z = -2t$

Line #2: $x = -1 + 2s$, $y = 3s$, $z = -7 + s$

For a P.O.I. the point (x, y, z) is the same for both lines. Equate the components.

x: $-1 + 3t = -1 + 2s \Rightarrow 3t - 2s = 0$ ①

y: $1 + 4t = 3s \Rightarrow 4t - 3s = -1$ ②

z: $-2t = -7 + s \Rightarrow -2t - s = -7$ ③

Pick two equations and solve for s & t. Then check values in 3rd equation.

① $3t - 2s = 0$

$-2 \times$ ② $-4t - 2s = -14$

$7t = 14$

$t = 2$

Sub $t = 2$ into ③: $-2(2) - s = -7$, $-4 - s = -7$, $s = 3$

Check in ②: $LS = 4t - 3s$, $RS = -1$

$= 4(2) - 3(3)$

$= 8 - 9$

$= -1$

$LS = RS$ ∴ "consistent"

P.O.I.

$x = -1 + 3(2) = 5$

$y = 1 + 4(2) = 9$

$z = -2(2) = -4$

∴ P.O.I. = $(5, 9, -4)$

The other option to solve is to write one equation in parametric form and sub into the other either in symmetric form (R^3) or Cartesian form (R^2)

same equations as example 1.

Determine the point of intersection: $\vec{r}_1 = (-1,1,0)+t(3,4,-2), t \in \Re$
$\vec{r}_2 = (-1,0,-7)+s(2,3,1), s \in \Re$

Line #1

$x = -1+3t$
$y = 1+4t$
$z = -2t$

Line #2

$\frac{x+1}{2} = \frac{y}{3} = \frac{z+7}{1}$

Sub L1 into L2.

$\frac{(-1+3t)+1}{2} = \frac{(1+4t)}{3}$ and $\frac{(1+4t)}{3} = \frac{(-2t)+7}{1}$

$9t = 2+8t$ $\quad$ $1+4t = -6t+21$

$t = 2$ $\quad$ $10t = 20$

same "consistent" $\quad$ $t = 2$

$\therefore t = 2$.

using line #1. (t is the parameter)

$x = -1+3(2) = 5$
$y = 1+4(2) = 9$
$z = -2(2) = -4$

$\therefore$ P.O.I. $(5,9,-4)$

Example 2: Determine the point of intersection between the two lines:

$\vec{r}_1 = (1,2,3)+t(1,-1,2), t \in \Re$
$\vec{r}_2 = (3,1,4)+s(4,1,3), s \in \Re$

$\vec{m}_1 = (1,-1,2)$
$\vec{m}_2 = (4,1,3)$
} non-collinear

$\therefore$ intersecting or skew lines.

Line#1

$x = 1+t$
$y = 2-t$
$z = 3+2t$

Line #2

$\frac{x-3}{4} = \frac{y-1}{1} = \frac{z-4}{3}$

we already checked for parallel.

Sub L1 into L2

$\frac{(1+t)-3}{4} = \frac{(2-t)-1}{1}$ and $\frac{(2-t)-1}{1} = \frac{(3+2t)-4}{3}$

$t-2 = 4-4t$ $\quad$ $3-3t = -1+2t$

$5t = 6$ $\quad$ $-5t = -4$

$t = \frac{6}{5}$ $\quad$ $t = \frac{4}{5}$

not same

$\therefore$ inconsistent

Since t-values not the same.
$\therefore$ no P.O.I.
$\therefore$ The two lines must be skew.

Example 3: Determine the point of intersection between the two lines:

$\vec{r}_1 = (-3,4)+t(2,-1), t \in \Re$
$\vec{r}_2 = (-7,1)+s(-2,1), s \in \Re$

$\vec{m}_1 = (2,-1)$
$\vec{m}_2 = (-2,1)$
} collinear

$\therefore$ parallel or coincident lines.

Option #1

Check to see if same line but checking a point from Line 1 into Line 2.

Line #1 point $(-3,4)$ into Line #2 parametric:

$x = -7+2s$ $\quad$ $y = 1+s$

$-3 = -7+2s$ $\quad$ $4 = 1+s$

$4 = 2s$ $\quad$ $3 = s$

$-2 = s$

not the same
$\therefore$ parallel & distinct!!

OR

Try to solve the system like usual.

Line #1

$x = -3+2t$
$y = 4-t$

Line #2

$\frac{x+7}{-2} = \frac{y-1}{1}$

Sub L1 into L2

$\frac{(-3+2t)+7}{-2} = \frac{(4-t)-1}{1}$

$2t+4 = -6+2t$

$0t = -10$.

not possible.

$\therefore$ No solutions
$\therefore$ must be parallel and distinct lines